NANYANG JUNIOR COLLEGE
Year 2 Preliminary Examination

H2 GEOGRAPHY 9173/02

Paper 2 21 Sept 2026

INSERT

3 hours

READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

Resource 1 for Question 1

Photographs students took of business premises and green facilities.









Sources:

https://www.aswatson.com/latest-news/a-s-watson-greener-stores-accelerate-global-movement-towards-a-more-sustainable-future

https://www.ntu.edu.sg/ntupreneur/our-students-and-alumni/stories/tour-to-the-hmgisc

https://robinsonsmalls.com/EarthDay2025

https://sbr.com.sg/building-engineering/exclusive/surbana-jurongs-campus-in-nature-transcends-sustainable-innovation-scale-employee-productivity

Resource 2 for Question 1

Survey questionnaire administered to TNC and SME representatives

Purpose Statement:
Hi! We are a group of students doing a Geography project. Please answer our survey about the environmental habits in your company/ shop. Thank you for your time.

SECTION A: BACKGROUND INFORMATION

1. Type of Business Operation:

- [ ] Local Shop / SME Tenant
- [ ] Global Corporation / TNC Office

2. What is your firm's annual sustainability and green investment budget?

- [ ] Less than $5,000
- [ ] $5,001 – $20,000
- [ ] More than $20,000

SECTION B: CLIMATE ACTION & MITIGATION MEASURES

3. Please indicate your level of agreement with the following statement:
"Our business actively prioritizes operational choices that address global warming and climate change."

- [ ] Strongly Disagree
- [ ] Disagree
- [ ] Neutral
- [ ] Agree
- [ ] Strongly Agree

4. How often do you set your shop's air-conditioning system to 23°C or below during operating hours?

- [ ] Everyday
- [ ] 3–4 times a week
- [ ] Seldom
- [ ] Never

5. Which packaging model best describes your store's regular operational practice?

- [ ] Primary reliance on single-use plastic bags / packaging materials
- [ ] Complete integration of zero-waste or biodegradable alternatives

6. Are you actively auditing your suppliers' carbon footprints?

- [ ] YES
- [ ] NO

THANK YOU FOR COMPLETING THE QUESTIONNAIRE!

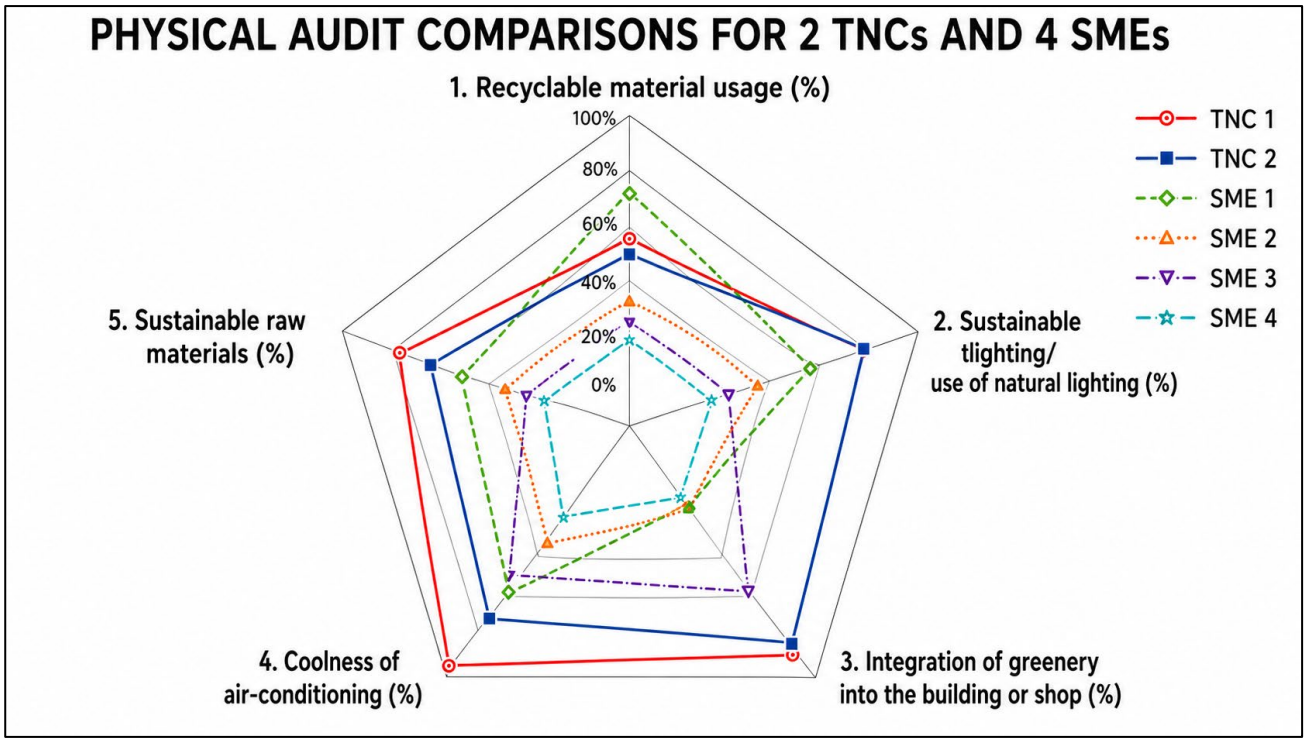
Resource 3 for Question 1

Data compiled from the questionnaire survey

Questions	Answers				
Q3 - Our business actively prioritizes operational choices that address global warming and climate change	Strongly Disagree [0]	Disagree [0]	Neutral [0]	Agree [4]	Strongly Agree [2]
Q4 - How often do you set your shop's air-conditioning system to 23°C or below during operating hours?	Everyday [0]	3–4 times a week [3]	Seldom [3]	Never [0]	
Q5 - Which packaging model best describes your store's regular operational practice?	Primary reliance on single-use plastic bags / packaging materials [1]	Complete integration of zero-waste or biodegradable alternatives [5]			
Q6 - Are you actively auditing your suppliers' carbon footprints?	Yes [6]	No [0]			

Resource 4 for Question 1

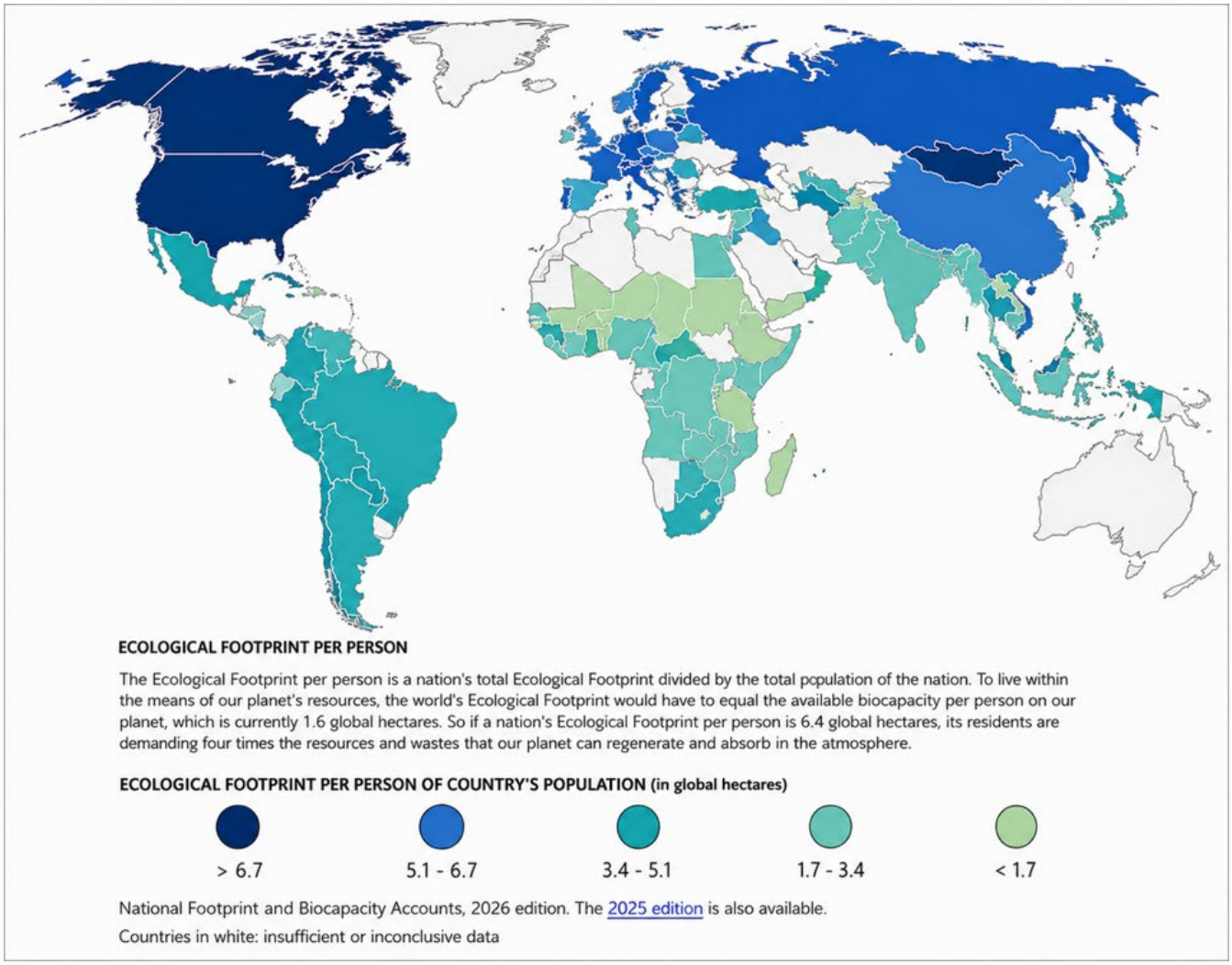
The physical audit of sustainability performance for the two TNCs and four SMEs



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Resource 5 for Question 2

Global distribution of ecological footprint per person (in global hectares)



Source: Adapted from National Footprint and Biocapacity Accounts, 2026 edition

Resource 6 for Question 2

Municipal Solid Waste (MSW) generation, recycling, and disposal by key material categories for the United States (based on 2018 EPA data)

Material Category	Total Generated (millions of metric tonnes)	Total Recycled / Composted (millions of metric tonnes)	Total Disposed (millions of metric tonnes)
Paper & Paperboard	61.1	41.7	19.4
Food Waste	57.2	2.4	54.9
Plastics	32.4	2.8	29.6
Yard Trimmings	32.1	20.2	11.9
Metals	23.2	7.9	15.3
Wood	16.4	2.8	13.6
Glass	11.2	2.8	8.3
Other (Textiles, Rubber)	31.6	4.5	27.0
Total MSW	**265.3**	**85.2 (32.1%)**	**180.1**

Note:
According to the U.S. Environmental Protection Agency (EPA), yard trimmings include grass, leaves, and tree and brush trimmings from residential, institutional and commercial sources.

Source: Adapted from United States Environmental Protection Agency (EPA)

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Resource 7 for Question 2

An excerpt from the United Nations report "The Weight of Cities: Resource Requirements of Future Urbanisation"

The proportion of the global population living in cities and towns is expected to rise from 54 per cent in 2015 to 66 per cent by 2050, which will result in a significant expansion of existing cities, as well as the construction of new cities. Without a new approach to urbanisation, material consumption by the world's cities will grow from 40 billion tonnes in 2010 to about 90 billion tonnes by 2050. Therefore, the resource use implications and environmental impacts of urbanisation are significant... Building and operating these new cities, and supporting the urban lifestyles of those who live in them, requires billions of tonnes of raw materials, such as fossil fuels, sand, gravel, iron ore, wood and food. Historically, existing cities have been spreading at a rate of two per cent a year, increasing global urban land use from just below one million square kilometres to 2.5 million in 2050, and putting agricultural land and food supplies at risk.

Source: United Nations Environment Programme (UNEP) - International Resource Panel.

Resource 8 for Question 2

An aerial photograph showing the scale of the Bantar Gebang landfill in Indonesia and its proximity to surrounding infrastructure and waterways



Source: https://www.reuters.com/business/environment/indonesia-says-7-killed-landfill-collapse-rescue-operation-ends-2026-03-10/

End of Paper